MDI Users Guide

The never-finished manual of the Micro Data Infrastructure

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2025-10-02

Contents

Overview of MDI **3**

Introduction to MDI **3**
Overview of MDI . . . 3
Stakeholders in the MDI Ecosystem . . . 3

Stakeholder NSI and Partners **4**
The MDI at an NSI site . . . 4
Importing to, and exporting from, the NSI site . . . 5
Initial set-up of MDI at the NSI (launch_v2.x) . . . 6
The directories at the NSI site MDI installation . . . 6
Files in the MDI folder . . . 6
subdirectories of `docs` (not shown here. MDI documentation) . . . 6
directory `launchpad` (with files to launch MDI) . . . 6
subdirectories of `rocket` (with code, and (meta)data to support MDI) . . . 6
subdirectories of `payload` (with analytical code and MD (meta)data) . . . 7
Preparing NSI metadata . . . 8
Launching the rocket and delivering output . . . 8
Mechanics of countdown, liftoff, rocket, payload . . . 9
Specifications for the NSI Metadata . . . 9
1. List of NSI dataFiles – *NSI*_datafiles.csv . . . 10
2. File-specific metadata – *NSI*_varnames.csv . . . 11
3. Codebook for categorical variables – *NSI*_codebook.csv . . . 13
4. Classification lists – *NSI*_*classvar*_class.csv . . . 14
5. Time concordances for classifications – *NSI*_*classvar*t0_*classvar*t1_timeconc.csv . . . 14
Specifications for the MD Metadata . . . 15
1. List of micro-dataset (MD) panels – MD_datafiles.csv . . . 15
2. Micro-dataset (MD) variables – MD_varnames.csv . . . 16
3. Classification lists – MD_*classvar*_class.csv . . . 18
4. Hierarchy files for classifications – MD_*classvar*_hier.csv . . . 18
5. Codebook for categorical variables – MD_codebook.csv . . . 19
Specifications for metadata needed for the NSI to MD harmonization . . . 23
1. Concordance file – *NSI*_MD_conc.csv . . . 23
2. *NSI*_MD_codeconc.csv . . . 25

Disclosure Criteria and Adherence **26**
What are Disclosure Criteria . . . 26
Components of Disclosure Criteria in the MDI: . . . 26
Disclosure Criteria in MDI Countries . . . 27
How Disclosure Criteria Are Applied: . . . 27
Disclosure Tools in MDI . . . 27

MDI Staff + Module Writers **28**
Setup for Researcher . . . 28
MDI Rtools . . . 28
Auxiliary Files . . . 28
Deflators . . . 28
Test Server and Mockdata Environment . . . 29

Module Writers (MDI Users) **29**
Overview of MDI . . . 29
Writing Modules for MDI Launcher . . . 29
Key Steps in Module Development . . . 30
Practical module writing . . . 30
Importing Data . . . 30
Manipulating Data . . . 30
Aggregating and exporting . . . 31
Analysis of MDI Module Output . . . 31
Stacking metadata output files . . . 31

Work at NSI **32**
Some more details of steps at NSI . . . 32

Preparation for each Launch **32**
The Harmonzied MD Longitudinal Firm Panels . . . 33

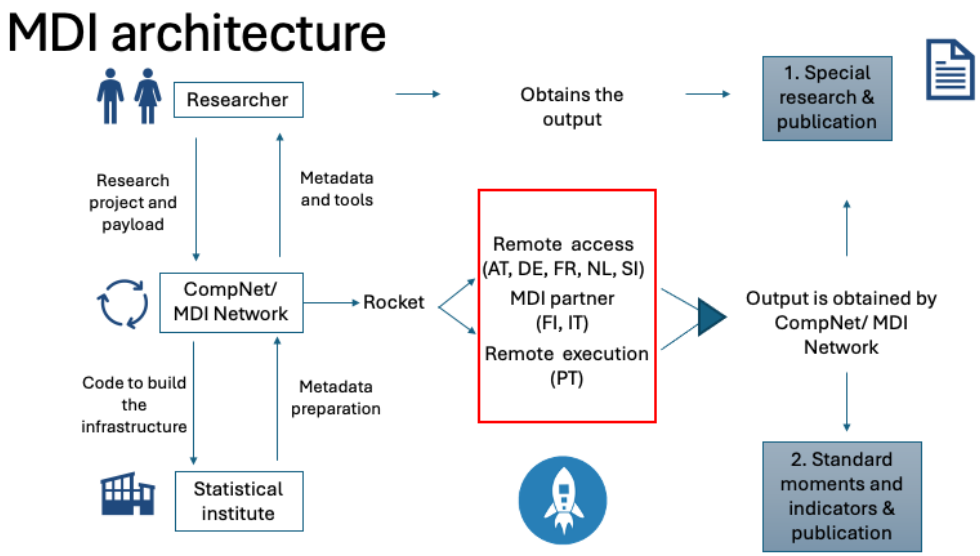
MDI Infra Tasks **33**
Preparation for Launch and Mission Control . . . 33
Pre-launch Testing . . . 33
Lockdown of version . . . 34
Interaction NSI - MDI during launch . . . 34
Work on MDI tools . . . 34

Mockdata **34**
break tasks into small progs (scripts) and tools (functions) . . . 35
Workflow for mock data simulation . . . 35

Nuvolos Server **36**
MDI Training Environment . . . 36
"Files" section . . . 36
Writing and executing code . . . 36
Internal Testing Environment . . . 36
First time users: How to connect Github . . . 37
The most common git commands . . . 37
names_select_Module . . . 38
mdi_mergeDTs() . . . 38
NSI_initialise() . . . 38

mdi_export() 38

Overview of MDI



1

Introduction to MDI

Overview of MDI

The Microdata Infrastructure (MDI) is designed with the dual goal of harmonizing firm-level data across countries and simplifying the research process for conducting cross-country analysis for a wide variety of research "themes". At its core, MDI offers a standardized environment where researchers can run identical analysis at multiple NSIs where the data are made comparable and accessible through a unified framework. The infrastructure supports a range of functionalities from data importation and harmonization to sophisticated analytical outputs, all within a controlled environment to maintain data confidentiality.

Stakeholders in the MDI Ecosystem

- National Statistical Institutes (NSIs) and other Partners
  - NSI remote execution
  - NSI remote access
  - Partners with country-specific (remote) access

NSIs provide the underlying data and support remote execution and/or access to confidential firm-level data. They vary in legal access rules, available data, and technical setups but are the backbone of the standardized MDI research environment.

- Module writers (MDI users)
  - Productivity Boards

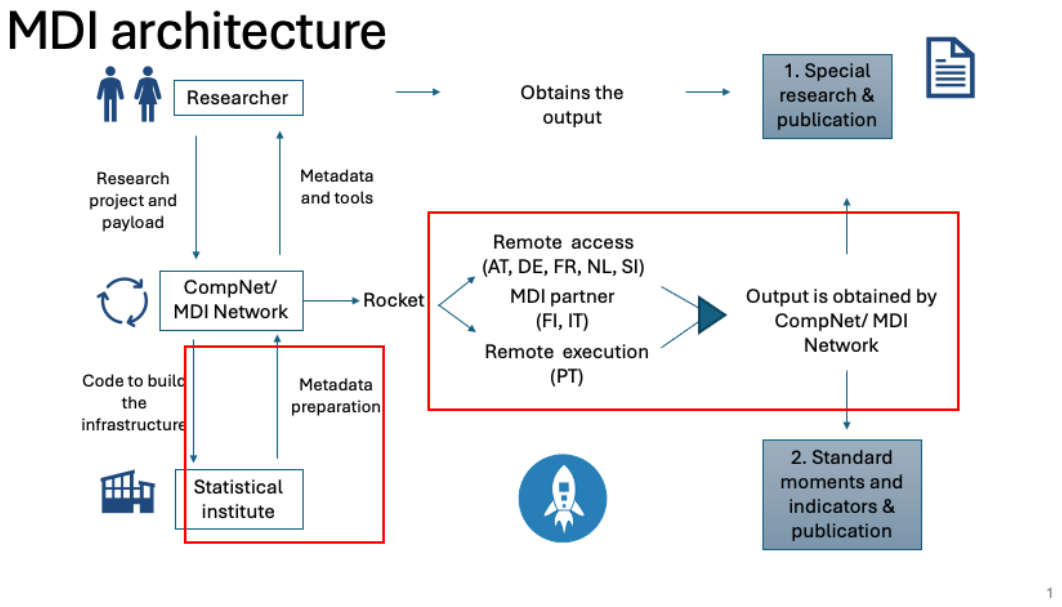
  - External Academic and Policy
  - MDI 'Theme' research staff

MDI Users includes productivity boards, external academic and policy researchers, and MDI 'Thematic' research staff. They are responsible for creating research modules that leverage MDI's capabilities for cross-country data analysis.

- MDI staff
  - Country specialists
  - Thematic research personnel
  - Infrastructure support teams

MDI staff ensure the development and smooth operation of the MDI environment, assist NSIs in data preparation and documentation, and assist module writers (MDI Users) with knowledge of data, tools, and research themes.

Stakeholder NSI and Partners



1

The MDI at an NSI site

- NSIs differ in legal access rules and technical setup for research use of firm-level data. The MDI facilitates access by providing a one-stop method for researchers to run their analysis as `payload` in an *MDI rocket* in multiple countries. MDI staff negotiates access contracts with each NSI according to national rules.
- NSIs differ in available data files (registers, surveys, administrative data sources) and specifications. The MDI provides researchers with a selection of micro-data (*MD*) panel datasets and variables that look identical across countries to computer code. The statistical and economic concepts are harmonized and the construction of the MD panels and variables from the underlying NSI sources are transparently documented.

- MDI metadata describe in each country the available files (surveys or register data), variables, and classification lists and are used to map all underlying data to the specification of harmonized MD firm-level panels
- At every NSI (including the `mock data site`), the MDI environment looks the same to the code in the `rocket` and its `payload`
- The rocket runs at the NSI in the MDI environment; reads data and metadata; and creates harmonized microdata (MD) panels.
- The 'payload' consists of research modules written in R with standardized tools and launch specific metadata. The payload code will generate statistical and analytical output in each country that can pass statistical disclosure. The output can therefore can be used for further cross-country analysis.

Importing to, and exporting from, the NSI site

- Whenever an import or export operation needs to be carried out from/to an environment, one should keep in mind how long it takes to do so and whether it involves a monetary cost
- The following table summarizes this information:

Country	Type	Time	Cost
NL	Import code/ metadata	1-2 days	free of charge
NL	Import data	1-2 days	free of charge
NL	Export code/ metadata	1-2 days	free of charge
NL	Light export results/ analysis	1-2 days	125 Euro
NL	Other export results/ analysis	1-2 days	250 Euro
SI	Import code/ metadata	1 day	free of charge
SI	Import data	1 day	free of charge
SI	Export code/ metadata	20-30 days	free of charge
SI	Light export results/ analysis	10-20 days	free of charge
SI	Other export results/ analysis	10 days	free of charge
FR	Import code/ metadata	1-2 days	free of charge
FR	Import data	1-2 days	free of charge
FR	Export code/ metadata	1-2 days	318 Euro for a 10 unit output package
FR	Light export results/ analysis	1-2 days	318 Euro for a 10 unit output package
FR	Other export results/ analysis	1-2 days	318 Euro for a 10 unit output package
PT	Import code/ metadata	few days	free of charge
PT	Light export results/ analysis	few days	free of charge
PT	Other export results/ analysis	few days	free of charge
FI	Import code/ metadata	2-3 days	free of charge
FI	Light export results/ analysis	2-3 days	free of charge
FI	Other export results/ analysis	2-3 days	free of charge
AT	Import files	21-45 days	most cases 1h * 118 Euro
AT	Export files	21-45 days	most cases 1h * 118 Euro

- In general, the files to be downloaded from MDI to the NSI, are stored in a TEAMS directory accessible to that NSI tema. The files are on the 'download' directory of a sharepoint file that you can share to your own machine, for example: "../OneDrive-SharedLibraries-IWHEconomicStudiesLab/MDI Data Providers Forum - PT"

- Each NSI also gets an MDI TEAMS directory 'upload' to upload output from the rocket.

Initial set-up of MDI at the NSI (launch_v2.x)

- For the *first time* using MDI, find a directory to place all the MDI files and subdirectories. The person setting up the MDI and running MDI `rockets` need to have read and write access to the directories and their subdirectories
- Install the four MDI directories–to be found on the MDI TEAMS cloud directory—under the main 'root' directory chosen for MDI:
  - docs (documentation of the MDI system, including this MDI Manual)
  - rocket (code to support and control MDI rocket launches, NSI metadata, auxilliary data)
  - payload (Research modules, includes metadata: NSI-specific NSI_MD concordances)
  - launchpad (NSI specific information to control MDI code and rocket launches)
- Make sure the user and the R programs have read access to the directory with raw data files
- Prepare a read/write directory for output files (pre-disclosure) create by the MDI R-code. After disclosure the files in this directory can be uploaded to the MDI TEAMS cloud directory available to the NSI staff.
- Fill in all the parameters needed in the file `launchpad/countdown.R`, including paths to relevant directories.

The directories at the NSI site MDI installation

Files in the MDI folder

```
.
+-- Infra
+-- MDI.Rproj
+-- MDI_github_commit.csv
+-- README.md
+-- alldocs
+-- docs
+-- launchpad
+-- mockdata
+-- output
+-- payload
\-- rocket
```

subdirectories of `docs` (not shown here. MDI documentation)

directory `launchpad` (with files to launch MDI)

```
launchpad
+-- README.md
+-- countdown.R
+-- interactive_MDI.R
+-- liftoff.R
+-- pre_launch_checker.R
+-- prepare_NSI.R
\-- report_file_changes.R
```

subdirectories of `rocket` (with code, and (meta)data to support MDI)

```
rocket
```

```
+-- CompNet
+-- MDIprogs
|   \-- metadata_tools
+-- NSImetadata
|   +-- AT
|   +-- DE
|   +-- FI
|   +-- FR
|   +-- NL
|   +-- Official
|   +-- PT
|   |   \-- Other
|   +-- PTx
|   \-- SI
+-- Rtools
|   +-- R
|   +-- Rpackages
|   \-- man
+-- auxdata
\-- control
```

subdirectories of `payload` (with analytical code and MD (meta)data)

```
payload
+-- Launch_v2.0
|   +-- MDmetadata
|   |   +-- DE
|   |   +-- NL
|   |   \-- SI
|   \-- Rmodules
|       \-- MP
+-- Launch_v2.1
|   +-- MDmetadata
|   |   +-- AT
|   |   +-- DE
|   |   +-- FI
|   |   +-- FR
|   |   +-- NL
|   |   +-- PT
|   |   \-- SI
|   \-- Rmodules
|       +-- CN
|       +-- EN
|       +-- FD
|       |   \-- auxdata
|       +-- MO
|       +-- MP
|       \-- TC
\-- Launch_v2.2
    +-- MDmetadata
    |   +-- DE
```

```
|   +-- FI
|   +-- FR
|   +-- NL
|   +-- PT
|   \-- SI
\-- Rmodules
    +-- CN
    +-- EN
    +-- FD
    |   \-- auxdata
    +-- MO
    +-- MP
    \-- TC
```

Preparing NSI metadata

- The NSI provides metadata of available firm-level data files.
- The NSI and MDI staff construct metadata to harmonize and link the raw files to the firm-level panel (*MD*) datasets, whose standards are described in the metadata of MD panels and variables provided by MDI.
- To aid in preparation of metadata, run `countdown.R` and choose program `prepare_nsi.R` (but not before filling out local information in the file `launchpad/countdown.R`). The preliminary metadata files are written to the dirOUTPUT directory and need to be editted. The instruction are given below in the section Metadata. The completed metadatafiles should be uploaed to the TEAMS directory and will, after processing and checking by the MDI team, be added to the NSImetadata directory for that country. The NSI can then download an updated version of the rocket, including the new metadata.

Launching the rocket and delivering output

- When a new launch date is set, the NSI needs to download the latest version of the docs and rocket directories from the MDI Teams cloud site (if there are new versions available), and overwrite the old versions.
- The new 'payload' subdirectory `launch_v2.x` directory needs to be downloaded and placed under the payload directory of the main MDI root directory.
- The file `countdown.R` should be copied to the user's directory and edited. Running `countdown.R` will allow user to choose further path of execution.
- Two folders need to created (with no constraints on how to name them – they can be stored to a directory of preference to the user):
  - Output folder
  - Folder for temporary files
- The following fields need to be manually specified, based on the directory system of the environment:
  - `dirMDI`: path to the MDI folder
  - `CountryCode`: 2-letter ISO 3166 code of the country of reference
  - `dirINPUTDATA`: path to the generic folder where the raw data is stored
  - `dirOUTPUT`: path to the output folder
  - `dirTMPSAVE`: path to the folder for temporary files
  - **Disclosure parameters**

    * `MDIminNumObs`: minimum number of firm-level observations underlying computation of cell of output file
    * `MDIdomSh`: maximum share of top X firms to meet dominance criterion
    * `MDIdomNr`: the X in top X firms.

- The program `pre_launch_checker.R` (run `countdown.R` and choose this program) needs to be run before anything else. It performs various checks on the NSI metadata to avoid errors later on. The results of the checks can be found in the file `pre_launch_checker_results.txt` in the output directory. It shows possible errors that should be adjusted in the NSI metadata. Additionally, two concordance files (NSI_pcc8t0_pcc8t1_conc.csv and NSI_MD_nace_conc.csv) are created using existing concordance files and updating them with the data at the NSI. These concordance table might contain empty values, if no value was previously defined. Missing values need to be filled in manually. When the concordance files are ready to be used, they need to be moved to the directory indicated in `pre_launch_checker_results.txt`
- The program `liftoff.R`, chosen after running `countdown.R`, will run the MDI rocket and analytical modules. Iterations with the MDI staff may be needed for fixes and patches to the rocket and payload, until the final results are written to the `dirOUTPUT` directory.
- The files in `dirOUTPUT` need to be checked for disclosure
- The files in `dirOUTPUT` can be uploaded to the MDI Teams upload directory.

Mechanics of countdown, liftoff, rocket, payload

- Copy `countdown.R` from MDI/launchpad/ to your own working directory. Fill in all parameters.
- Run `countdown.R` and choose which program to run:
- Run `pre_launch_checker.R` which you should run whenever a new version of the MDI system has been installed. It checks for consistency between NSI metadata known to MDI and the information in the data at the NSI site.
- Run `liftoff.R`, which calls control/rocket.R
  - `rocket.R` calls
    * load_MDIenvironment.R
    * document_output.R (generates information to document details of launch and output files)
    * read_MDmetadata.R
    * read_NSImetadata.R (EB: _datafiles, maybe others….)
    * read_auxdata.R (deflators for now, maybe WIOD, downstream, maybe these go in modules)
    * read_firmdata.R (based on selectnames for all modules….)
    * payload/launch_v2.x/run_modules.R executes the payload modules
    * cleanup.R cleans up the environment
- Run `prepare_NSI.R` which sets up the NSI environment to generate metadata
- Run `interactive_MDI` which you should run whenever working with the MDI system (other than liftoff or prepare).

Specifications for the NSI Metadata

This section summarizes the structure and content of the NSI metadata files. Specifically the metadata documents, in a machine and human readable form, the available data files, the description of the rows of the file (units of observation), the name and description of the variables (columns) in each file, the valid values (class and domain) for that variable in each row. The following paragraphs provide guidance on preparing the country specific metadata files.

Once created, the NSI metadata files need to be uploaded to the TEAMS directory. After the NSI downloads

the rocket, the metadata files of the NSI will be in the `rocket/NSImetadata/*NSI*/` directory of the MDI infrastructure. The MDI program called `pre_launch_checker.R`, to be run whenever MDI is updated, will flag inconsistencies and other problems in the metadata.

There are various types of (human- and machince-readable) metadata files prepared by the NSI: 1. A file that lists the available NSI firm-level datafiles (_datafiles) 2. Files that document the variables and their descriptions for each raw datafile listed above (_varnames). 3. Files that map categorical answers to their description (_codebook). 4. Files that describe classification variables on the datasets (_class, e.g. industry or product codes).

Together with the MDI team, the NSI also creates metadata to aid in harmonizing the NSI data to the MD specification. The MDI team provides, possibly specific to each launch, the metadata describing the MD datasets and its variables. Further, the MDI team, together with the NSI, provide concordances that are used to harmonize the NSI data files to the common format.

In the filenames for the metadata, the acronym *NSI* is used. This should be substituted with the 2-letter country code for the country in question (using the ISO3166-2 standard, e.g. `NSI = PT`). For the MDI metadata, the two letters `MD` are used, and for the `mock data` test site, the letters `TS` are used.

1. List of NSI dataFiles – *NSI*_datafiles.csv

This file contains the list of all available raw data files on a country's environment. The file has the following columns:

`[1] NSI_dataset,NSI_datafile,yearvar,year_start,year_end,format,path,details`

where,

- `NSI_dataset` is the 'generic' name of the NSI datafile
- `NSI_datafile` is the name of the file in the NSI environment
- `yearvar` gives name of year variable if `NSI_datafile` is a panel, empty otherwise
- `year_start` is the starting year of the data file
- `year_end` is the last year of the data file (if empty or NA, the file is a panel and includes a variable 'year')
- `format` is the file extension (csv, sas, stata, etc) of the file (i.e. also the storage format of the data)
- `path` indicates path of the datafile relative to the NSI data directory (given by the parameter `dirINPUTDATA` in launchpad/countdown.R)
- `details` contains additional notes on the file

An example (for NSI=FI, 2018) of the metadata for the raw data files (the columns `yearvar`, `year_end`, `path` and `details` are omitted for viewing):

NSI_dataset	NSI_datafile	year_start	format
bd	bd2018	2018	csv
br	br2018	2018	csv
bs	bs2018	2018	csv
cis	cis2018	2018	csv
ifats	ifats2018	2018	csv
itgs	itgs2018	2018	csv
its	its2018	2018	csv
ofats	ofats2018	2018	csv
prodcom	prodcom2018	2018	csv

sbs	sbs2018	2018	csv
ictec	ictec2018	2018	csv

2. File-specific metadata – *NSI*_varnames.csv

This file contain the list of all variables in each raw datafile appearing in the column `NSI_datafile` of *NSI*_datafiles.csv. Each file has the following columns:

`[1] NSI_datafile,NSI_varname,is_key,description,class,domain`

where

- `NSI_datafile` is the name of the file in the NSI environment
- `NSI_varname` is the name (hopefully mnemonic) of the variable in the raw file
- `is_key` is a boolean stating whether variable belongs to the (possibly joint) unique keys of the datafile, e.g. `firmid`, or `firmid,year` are often the unique key(s).
- `description` contains a description of the variable, if possible using Eurostat convention
- `class` is the type of value that the variable holds (e.g. numeric, date, character, logical)
- `domain` provides information on the values of the variable. See examples below:
  - classification: e.g. list of industry, region, product or codes. (values is metadata filename: e.g. *NSI_classname*_class.csv, which provides a list of permissible values and descriptions)
  - file-specific codebook of categorical answers. (value is metadata filename, e.g. *NSI_codebook.csv containing permissible values, such as 'yes', 'no','maybe', or 'small', 'large', 'medium'.
  - For other values:
    - For monetary values, "1000" (for 1000 Euros)
    - For dates: "%m%d%Y" (R date-format for mmddyyyy). For 'year' variable, we use "%Y"
    - For real units, choose from: "ton" (weight, 1000kg), "m3" (volume), "GJ" (energy), "unit" (1 item).

Domain: Expenditures, Quantities, Dates

Measure	Domain Entry	Description
Expenditure	1000	… or 1 Euro; 10000000 Euro; etc.
Foreign currency	1*FXC	… or 1000 etc.; Where FXC is an ISO 4217 3-letter currency code
Employment	1	1 here refers to 1 FTE; or 1000; … or 1 Emp if in persons
Numerical	1	1 here refers to 1 unit; … or 10; 100; where 'unit' gives unit in lowercase for the variable in the NSI data file
Date	%Y-%m-%d	Use the R date format that matches the values for the NSI date or year variable

Here is an abbreviated list of R date formats:

Format	Description	Example
%a	Abbreviated weekday	Sun, Thu
%A	Full weekday	Sunday, Thursday
%b or %h	Abbreviated month	May, Jul
%B	Full month	May, July

Format	Description	Example
%d	Day of the month 01-31	27, 07
%j	Day of the year 001-366	148, 188
%m	Month 01-12	05, 07
%U	Week 01-53, (start Sunday)	22, 27
%w	Weekday 0-6 (Sunday= 0)	0, 4
%W	Week 00-53 (start Monday)	21, 27
%x	Date, locale-specific	
%y	Year 2-digit 00-99	84, 05
%Y	Year 4-dig: (69 to 99 - 19xx)	1984, 2005
%C	Century	19, 20
%D	Date formatted %m/%d/%y	05/27/84, 07/07/05
%u	Weekday 1-7 (Monday=1	7, 4

Domain: Classification or Categorical (factor) variables

Variable	Domain_Entry	Description
Classification variable	NSI_classname_cas	As (official) list, ie NL_nace
Categorical variable	NSI_codebook	Contains permissible values for categorical (factor) variables, e.g. 'yes', 'no','maybe'

Example: Netherlands (SBS, 2018): NL_varnames.

NSI_datafile	NSI_varname	is_key	description	class	domain
sbs2018	ent_id	1	Enterprise ID (identification	character	
sbs2018	sbs_12110	0	Turnover	numeric	1000
sbs2018	sbs_12150	0	Value added at factor cost	numeric	1000
sbs2018	sbs_12170	0	Gross operating surplus	numeric	1000
sbs2018	sbs_13110	0	Total purchases of goods and s	numeric	1000
sbs2018	sbs_13310	0	Personnel costs	numeric	1000
sbs2018	sbs_13320	0	Wages and salaries	numeric	1000
sbs2018	sbs_15110	0	Gross investment in tangible g	numeric	1000
sbs2018	sbs_15150	0	Gross investment in machinery	numeric	1000
sbs2018	sbs_15429	0	Gross investment in concession	numeric	1000
sbs2018	sbs_15441	0	Investment in purchased softwa	numeric	1000
sbs2018	sbs_16110	0	Number of persons employed	numeric	1
sbs2018	sbs_16130	0	Number of employees	numeric	1
sbs2018	sbs_16140	0	Number of employees in full-ti	numeric	1
sbs2018	sbs_20110	0	Purchases of energy products (	numeric	1000
sbs2018	sbs_22110	0	Total intra-mural R & D expend	numeric	1000
sbs2018	sbs_22120	0	Total number of R & D personne	numeric	1000

sbs2018	sbs_type	0	Code to show if data (unit) is	numeric	NL_codebook

3. Codebook for categorical variables – *NSI*_codebook.csv

This file contains the possible values of a categorical variable and the description that belongs to that value. There rows give the possible values occuring in firm data for a particular `NSI_datailfe` and `NSI_varname`. The name of the codebook should be given in the 'domain' columnn of NSI_varnames for the relevant categorical variable.

`[1] NSI_dataset,NSI_varname,year,code,description`

[`NSI_datafile`,`NSI_varname`, `year`, `code` , `description`]

where,

- `NSI_dataset` is the name of the generic dataset in the NSI environment
- `NSI_varname` is the name of the variable of that specific raw dataset
- `year` is the year for which codebook values hold. If empty, holds for all years of the NSI dataset
- `code` gives all the values of the categorical variable that occur for that `NSI_varname` in that `NSI_datafile`
- `description` gives the description explaining each code value

An example below of the values occuring for the unit of measurement in the SI `ENER` data for 2012.

NSI_dataset	year	NSI_varname	code	description
MIKRO_INDL_razST	NA	ME	1000 SIT	thousands of slovenian tolars
MIKRO_INDL_razST	NA	ME	EUR	euros
MIKRO_INDL_razST	NA	ME	GJ	Gigajoule - a unit of energy
MIKRO_INDL_razST	NA	ME	MWh	Megawatt-hour - a unit of energy
MIKRO_INDL_razST	NA	ME	TJ	terajoules
MIKRO_INDL_razST	NA	ME	kW	kilowatts
MIKRO_INDL_razST	NA	ME	kg	Kilogram - a unit of mass
MIKRO_INDL_razST	NA	ME	kg N	kilograms of nitrogen
MIKRO_INDL_razST	NA	ME	kg P2O5	Kilogram of Phosphorus Pentoxide - a mea
MIKRO_INDL_razST	NA	ME	kg akt. subst.	Kilogram Active Substance - a measure of
MIKRO_INDL_razST	NA	ME	kg akt.subst	Kilogram Active Substance - a measure of
MIKRO_INDL_razST	NA	ME	kg efekt.	Kilogram Effective - likely a measure of eff
MIKRO_INDL_razST	NA	ME	kos	Cost - possibly referring to cost in monetar
MIKRO_INDL_razST	NA	ME	l	Liter - a unit of volume
MIKRO_INDL_razST	NA	ME	l alc 100%	Liter Alcohol 100% - pure alcohol volume
MIKRO_INDL_razST	NA	ME	m	Meter - a unit of length
MIKRO_INDL_razST	NA	ME	m2	Square Meter - a unit of area
MIKRO_INDL_razST	NA	ME	m3	Cubic Meter - a unit of volume
MIKRO_INDL_razST	NA	ME	par	Pair - possibly referring to a set of two iter
MIKRO_INDL_razST	NA	ME	t	Ton - a unit of mass
MIKRO_INDL_razST	NA	ME	tiso_ SIT	thousands of slovenian tolars
MIKRO_INDL_razST	NA	ME	tiso_ ef.ur	Thousand Effective Hours - likely referring

MIKRO_INDL_razST	NA	ME	tisoč SIT	Thousand Slovenian Tolars - currency unit
MIKRO_INDL_razST	NA	ME	tisoč ef.ur	Thousand Effective Hours - likely referring

4. Classification lists – *NSI_classvar*_class.csv

This file contains the unique list of codes per year of a specific classification variable in a country. Note that there should be a list for every categorical variable in each dataset. The related table has the following columns:

```
[1] code,year,description
```

where,

- `code` is the list values of the classification variable observed in the data
- `year` is the related year
- `description` gives the description for each code value

An sample of rows from the table of NACE codes (in this case the official EU NaceR2 classification):

code	description
C10	___Manufacture of food products
C10.1	_____Processing and preserving of meat and production of meat products
C10.1.1	_____Processing and preserving of meat
C10.1.2	_____Processing and preserving of poultry meat
C10.1.3	_____Production of meat and poultry meat products
C10.2	_____Processing and preserving of fish, crustaceans and molluscs
C10.2.0	_____Processing and preserving of fish, crustaceans and molluscs
C10.3	_____Processing and preserving of fruit and vegetables
C10.3.1	_____Processing and preserving of potatoes
C10.3.2	_____Manufacture of fruit and vegetable juice
C10.3.9	_____Other processing and preserving of fruit and vegetables
C10.4	_____Manufacture of vegetable and animal oils and fats
C10.4.1	_____Manufacture of oils and fats
C10.4.2	_____Manufacture of margarine and similar edible fats
C10.5	_____Manufacture of dairy products
C10.5.1	_____Operation of dairies and cheese making
C10.5.2	_____Manufacture of ice cream
C10.6	_____Manufacture of grain mill products, starches and starch products
C10.6.1	_____Manufacture of grain mill products
C10.6.2	_____Manufacture of starches and starch products

5. Time concordances for classifications – *NSI_classvar*t0_*classvar*t1_timeconc.csv

This file contains the a concordance table for classification lists between a code at time `t-1` and the corresponding code at `t`. The reference year is indicated in column `year`. In other words, `t=year`.

```
[1] left,right,year
```

where,

- `left` is the code at time `t-1`
- `right` is the value(s) the code in column `left` can take at time `t`
- `year` is the reference year

For example, the following table displays how such table looks like for a subset of ITGS codes in PT:

left	right	year
85178190	85178090	1995
9619 00 13	9619 00 75	2013
90172039	90172039	1997
73121095	73121084	1994
48149090	48149080	2006
6815 10 10	6815 13 00	2021
9093090	9093000	2000
49111090	49111000	1989
4407 99 27	4407 29 96	2016
85013110	85013100	2005
55039010	55039000	2008
38069000	38062000	1995
85421991	85423091	1995
4407 10 91	4407 12 90	2016
28342930	28520000	2006

Specifications for the MD Metadata

- Work in this section is a collaboration between NSI and MDI staff
- In an iterative process, using NSI metadata for each country, and taking into account research needs of MD users, a specification is made of the MD panels and variables.
  1. MD_datafiles.csv describe the harmonized panel datasets generated in each launch
  2. MD_varnames.csv describe the variables per dataset, with their description, class, and domain.
  3. MD classifications: versions of official classifications, such as EU NaceR2 activities or NUTS2 regions
  4. MD codebooks: valid values for categorical variables

1. List of micro-dataset (MD) panels – MD_datafiles.csv

This file contains the list of all firm-level MD panels generated by the MDI code and usable by researchers via code modules. The file has following columns:

This file contains the list of all harmonized firm-level micro data panels (MD datafiles) that can be used in research by code in an MDI launch, either individually, or linked at the firm-year level.

`[1] MD_dataset,description,details`

where,

- `MD_dataset` is the name of the MD panel (R data.table) at runtime of the launch
- `description` A description of the panel and its underlying source data

- `details` contains additional notes on the file

Below is the list of currently available MD panels:

MD_dataset	description	details
BR	Business Register	see: https://ec.europa.eu/eurostat/
BS	Balance Sheet	Balance Sheet on Enterprise groups
BD	Business Dynamics	
SBS	Structural Business Statistics	
CIS	Community Innovation Survey	(only available in even numbered ye
ICTEC	ICT Usage in Enterprises Survey	https://ec.europa.eu/eurostat/cache
ITGS	International Trade in Goods	
ITS	International Trade in Services	
OFATS	Outgoing Foreign Affiliates Statist	
IFATS	Incoming Foreign Affiliates Statsti	
ENER	Energy Use at Firms	in progress harmonization across co
PRODCOM	Production Communitaire by firm and	https://ec.europa.eu/eurostat/web/p

2. Micro-dataset (MD) variables – MD_varnames.csv

This file contains the list of all variables available in all the MD firm-level panel datsets that have been generated by the MDI code using the NSI datafiles, NSI metadata, and the NSI-MD concordances. The file has the following columns:

`[1] MD_varname,MD_dataset,is_key,description,class,domain`

where

- `MD_dataset` is the name of the MD firm-level panel dataset, ie BR, SBS, etc.
- `MD_varname` is the name of the variable in the virtual firm-level dataset
- `is_key` is a boolean stating whether variable belongs to the (possibly joint) unique keys of the dataset, e.g. `firmid`, or `firmid,year` are often the unique key(s).
- `description` contains a description of the variable, if possible using Eurostat convention
- `class` is the type of value that the variable holds (e.g. integer, character, boolean etc.)
- `domain`
  - classification: e.g. list of industry, region, product or codes. (values is metadata filename: e.g. *MD_filename_varname_*list.csv, which provides a list of permissible values and descriptions)
  - MD-specific codebook of categorical answers. (value is metadata filename, e.g. *MD_codebookname_*codes.csv containing permissible values, such as 'yes', 'no','maybe'
  - For other values:
    - For monetary values, "1000" (for 1000 Euros)
    - For dates: "%m%d%Y" (R date-format for mmddyyyy). For 'year' variable, we use "%Y"
    - For real units, choose from: "ton" (weight, 1000kg), "m3" (volume), "GJ" (energy), "unit" (1 item).

Domain: Expenditures, Quantities, Dates

Measure	Domain_Entry	Description
Expenditure	1000 Euro	
Employment	1 FTE	… or 1 Emp if in persons
Numerical	1 'unit'	'Unit' gives unit used in NSI data file, or is left blank if just a count.
Date	%Y	For now, we use a R format for 4-digit year as the date variable
Weight	1 kg	
Volume	1 m3	
Area	1 m2	
Length	1 m	
Energy	1 GJ	GigaJoule

Domain: Classification or Categorical (factor) variables

Variable	Domain_Entry	Description
Classification variable	NSI_classname_cass	Any (official) list, ie NL_nace
Categorical variable	NSI_codebook	Contains permissible values for categorical (factor) variables, e.g. 'yes', 'no','maybe'

Below is a random sample of 20 rows of the file MD_varnames with harmonized MD variables

MD_dataset	MD_varname	description	domain
ICTEC	INTRA	Firm has intranet	
SBS	ni_sw	Investment in purchased s	1000
BS	year	Year	%Y
ICTEC	CRMAN	analyse information for m	
BR	firmid	Unique enterprise identif	
BS	fixed_asse	non-current assets :tangi	1000
SBS	persons	Number of persons employe	1
SBS	nm	Total purchases of goods	1000
SBS	emp	Number of employees	1
BS	other_curr	current assets not subsum	
ICTEC	IACC	Firm has internet	
ITGS	ctry	Partner country (country	MDiso3166-2_class
SBS	nv	Value added at factor cos	1000
BS	current_as	Current assets according	1000
BS	equity	total shareholders' funds	1000
ICTEC	ADE	Automated Data Exchange	
ITS	imputed	Code to show if data (uni	
SBS	wages	Wages and salaries	1000
SBS	ni_tan	Gross investment in tangi	1000
ICTEC	AEBVALPCT	% of orders through inter	MD_codebook

3. Classification lists – MD__*classvar*__class.csv

This file contains the unique list of codes per year of a specific classification variable from the MD panels. Note that there should be a list for every categorical variable in each MD datasets. The related table has the following columns:

[`code` , `description`]

where,

- `code` is the list values of the classification variable observed in the data
- `description` gives the description for each code value

An example of the table for NACE codes (in this case the official EU NaceR2 classification):

code	description
C17.2.2	_____Manufacture of household and sanitary goods and of toilet requisit
C17.2.3	_____Manufacture of paper stationery
C17.2.4	_____Manufacture of wallpaper
C17.2.9	_____Manufacture of other articles of paper and paperboard
C18	___Printing and reproduction of recorded media
C18.1	____Printing and service activities related to printing
C18.1.1	_____Printing of newspapers
C18.1.2	_____Other printing
C18.1.3	_____Pre-press and pre-media services
C18.1.4	_____Binding and related services
C18.2	____Reproduction of recorded media
C18.2.0	_____Reproduction of recorded media
C19	___Manufacture of coke and refined petroleum products
C19.1	____Manufacture of coke oven products
C19.1.0	_____Manufacture of coke oven products
C19.2	____Manufacture of refined petroleum products
C19.2.0	_____Manufacture of refined petroleum products
C20	___Manufacture of chemicals and chemical products
C20.1	____Manufacture of basic chemicals, fertilisers and nitrogen compounds,
C20.1.1	_____Manufacture of industrial gases

4. Hierarchy files for classifications – MD__*classvar*__hier.csv

This file contains a series of columns that refer to different nodes of the classification variable in question. With this file, the user can easily aggregate or disaggregate the data based on the different nodes of the classification variable.

The columns of the file are labelled as `h_`X, where X is a number from 0 to N denoting one of the N available nodes in the variable.

An example of a hierarchy table for NACE codes (in this case the official EU NaceR2 classification):

h_0	h_1	h_2	h_3	h_4
1711	171	17	C	TOT
2815	281	28	C	TOT
3319	331	33	C	TOT
4939	493	49	H	TOT
220	22	2	A	TOT
4619	461	46	G	TOT
1610	161	16	C	TOT
4774	477	47	G	TOT
4791	479	47	G	TOT
4621	462	46	G	TOT
6190	619	61	J	TOT
5122	512	51	H	TOT
2342	234	23	C	TOT
311	31	3	A	TOT
8622	862	86	Q	TOT

5. Codebook for categorical variables – MD_codebook.csv

This file contains the possible values of a categorical variable and the description that belongs to that value. Note that sometimes a particular codebook is 're-used' for multiple variables. The name of the codebook should be given in the 'domain' columnn of the metadata for the file containing the categorical variable.

[`MD_dataset`,`MD_varname`, `code` , `description`]

where,

- `MD_dataset` is the name of the MD firm-level panel dataset, ie BR, SBS, etc.
- `MD_varname` is the name of the variable of that specific MD dataset
- `code` gives the valid values of the ccategorical variable
- `description` gives the description for each code value

An example below of the values given for some variables in the MD `br` business register dataset

MD_dataset	MD_varname	code	description
BD	status	1	born in reference year
BD	status	2	active entire reference year
BD	status	3	dead in reference year
BD	status	4	born and dead in reference year
BR	demo	0	"No demographic relation in ref. year"
BR	demo	1	"Receiving employment from other enterprise in ref. ye
BR	demo	2	"Transfers employment to other enterprise and cease to
BR	demo	3	" Transfers employment to other enterprise and continu
BR	lfo	LL	Limited liability company - include limited liability par
BR	lfo	SP	Sole Proprietor

BR	lfo	PA	Partnership - exclude limited liability partnerships
BR	lfo	GO	Government - local and central government - exclude p
BR	lfo	NB	Non profit body or mutual association
BR	lfo	NP	Natural person(s) - if economically active, code as sole
BR	lfo	ND	Not defined
BR	soe	1	private
BR	soe	2	public
BR	soe	9	not available
BR	soe		missing
ICTEC	aebvalpct	1	Less than 1%
ICTEC	aebvalpct	2	1% or more and less than 5%
ICTEC	aebvalpct	3	5% or more and less than 10%
ICTEC	aebvalpct	4	10% or more and less than 25%
ICTEC	aebvalpct	5	25% or more and less than 50%
ICTEC	aebvalpct	6	50% or more and less than 75%
ICTEC	aebvalpct	7	75% or more
ICTEC	aebvalpct	9	not applicable
ICTEC	aebvalpct		no answer
ITGS	ctrygrp	1	EU-15
ITGS	ctrygrp	2	EU-13
ITGS	ctrygrp	3	Other Europe
ITGS	ctrygrp	4	Russia
ITGS	ctrygrp	5	China
ITGS	ctrygrp	6	Japan
ITGS	ctrygrp	7	India
ITGS	ctrygrp	8	Brazil
ITGS	ctrygrp	9	USA
ITGS	ctrygrp	10	Canada & Mexico
ITGS	ctrygrp	11	Oceania & Rest of Asia
ITGS	ctrygrp	12	Central Am. & Rest of South Am.
ITGS	ctrygrp	13	Africa
ITGS	ctrygrp	14	Unknown
ITGS	ctrygrp	15	Other
ITS	ctrygrp	1	EU-15
ITS	ctrygrp	2	EU-13
ITS	ctrygrp	3	Other Europe
ITS	ctrygrp	4	Russia
ITS	ctrygrp	5	China
ITS	ctrygrp	6	Japan
ITS	ctrygrp	7	India
ITS	ctrygrp	8	Brazil
ITS	ctrygrp	9	USA

ITS	ctrygrp	10	Canada & Mexico
ITS	ctrygrp	11	Oceania & Rest of Asia
ITS	ctrygrp	12	Central Am. & Rest of South Am.
ITS	ctrygrp	13	Africa
ITS	ctrygrp	14	Unknown
ITS	ctrygrp	15	Other
OFATS	ctrygrp	1	EU-15
OFATS	ctrygrp	2	EU-13
OFATS	ctrygrp	3	Other Europe
OFATS	ctrygrp	4	Russia
OFATS	ctrygrp	5	China
OFATS	ctrygrp	6	Japan
OFATS	ctrygrp	7	India
OFATS	ctrygrp	8	Brazil
OFATS	ctrygrp	9	USA
OFATS	ctrygrp	10	Canada & Mexico
OFATS	ctrygrp	11	Oceania & Rest of Asia
OFATS	ctrygrp	12	Central Am. & Rest of South Am.
OFATS	ctrygrp	13	Africa
OFATS	ctrygrp	14	Unknown
OFATS	ctrygrp	15	Other
ITS	bus	10	Distribution, logistics
ITS	bus	11	Marketing, sales, etc.
ITS	bus	12	ICT and telecom.
ITS	bus	13	Adm., management
ITS	bus	14	R&D, engineering
ITS	bus	15	Other support functions
ITS	bus	16	Royalties and licens fees
ITS	bus	17	Other
PRODCOM	unit	1000 EUR	1000 euro
PRODCOM	unit	1 GJ	Gigajoule - a unit of energy
PRODCOM	unit	1 MWh	Megawatt-hour - a unit of energy
PRODCOM	unit	1 TJ	terajoules
PRODCOM	unit	1 kW	kilowatts
PRODCOM	unit	1 kg	Kilogram - a unit of mass
PRODCOM	unit	1 kg N	kilograms of nitrogen
PRODCOM	unit	1 kg P2O5	Kilogram of Phosphorus Pentoxide - a measure for fert
PRODCOM	unit	1 kg act. subst.	Kilogram Active Substance - a measure of active ingred
PRODCOM	unit	1 kg efekt.	Kilogram Effective - likely a measure of effective substa
PRODCOM	unit	1 l	Liter - a unit of volume
PRODCOM	unit	1 l alc 100%	Liter Alcohol 100% - pure alcohol volume
PRODCOM	unit	1 m	Meter - a unit of length

PRODCOM	unit	1 m2	Square Meter - a unit of area
PRODCOM	unit	1 m3	Cubic Meter - a unit of volume
PRODCOM	unit	1 pair	Pair - possibly referring to a set of two items
PRODCOM	unit	1 t	Ton - a unit of mass
PRODCOM	unit	1000 eff. work hrs.	Thousand Effective Hours - likely referring to time or v
PRODCOM	unit	1 piece	Number of pieces
PRODCOM	unit	1 gWh	Gigawatt-hour
PRODCOM	unit	1 kWh	Kilowatt-hour
PRODCOM	unit	gross tonnage	gross tonnage
PRODCOM	unit	1 km	Kilometres
PRODCOM	unit	1 m linear	Linear metres
PRODCOM	unit	number of steps	Number of steps
PRODCOM	unit	1 piece	pieces
PRODCOM	unit	1 paper roll	Paper rolls
PRODCOM	unit	number	Number
PRODCOM	unit	1000 pieces	Thousands of pieces
PRODCOM	unit	1 carat	carats
PRODCOM	unit	1000 m2	Thousands of squared metres
PRODCOM	unit	1000 m3	Thousands of cubic metres
PRODCOM	unit	1 ml	Millilitres
PRODCOM	unit	1 t SiO2	Tons of silicon dioxide
PRODCOM	unit	1 kg 90% dry	Kilogram of substance 90% dry
PRODCOM	unit	1 Kg H2SO4	Kilogram of sulfuric acid
PRODCOM	unit	1 g	Grams
PRODCOM	unit	1 t N	Tons of nitrogen
PRODCOM	unit	1 t K2O	Tons of potassium oxide
PRODCOM	unit	1 t P2O5	Tons of diphosphorus pentoxide
PRODCOM	unit	1 t TiO2	Tons of titanium dioxide
PRODCOM	unit	1 t Al2O3	Tons of dialuminium trioxide
PRODCOM	unit	1 t B2O3	Tons of diboron trioxide
PRODCOM	unit	1 t Cl	Tons of chlorine
PRODCOM	unit	1 t F	Tons of fluorine
PRODCOM	unit	1 t HCl	Tons of hydrochloric acid
PRODCOM	unit	1 t HF	Tons of hydrofluoric acid
PRODCOM	unit	1 t H2O2	Tons of hydrogen peroxide
PRODCOM	unit	1 t KOH	Tons of potassium hydroxide (caustic potash)
PRODCOM	unit	1 t NaOH	Tons of sodium hydroxide (caustic soda)
PRODCOM	unit	1 t Na2CO3	Tons of sodium carbonate
PRODCOM	unit	1 t Na2S2O5	Tons of sodium pyrosulphide
PRODCOM	unit	1 t SO2	Tons of sulphur dioxide
PRODCOM	unit	1000 kWh	Thousand kilowatt hours
PRODCOM	unit	ce/el	Number of cells

PRODCOM	unit	CGT	Compensated Gross Tonnes
PRODCOM	unit	kg Cl	Kilogram of chlorine
PRODCOM	unit	kg H2O2	Kilogram of hydrogen peroxide
PRODCOM	unit	kg HCl	Kilogram of hydrogen chloride
PRODCOM	unit	kg K2O	Kilogram of potassium oxide
PRODCOM	unit	kg KOH	Kilogram of potassium hydroxide (caustic potash)
PRODCOM	unit	kg Na2CO3	Kilogram of sodium carbonate
PRODCOM	unit	kg Na2S2O5	Kilogram of sodium metabisulfite
PRODCOM	unit	kg NaOH	Kilogram of sodium hydroxide (caustic soda)
PRODCOM	unit	kg SiO2	Kilogram of silicon dioxide (silica)
PRODCOM	unit	kg TiO2	Kilogram of titanium dioxide

Specifications for metadata needed for the NSI to MD harmonization

- Harmonization of MD panels entails harmonization of units of observation, variable definitions, and variable values
- The key to harmonization is NSI metadata, MD metadata, and NSI to MDI concordances
- The MD standard metadata is found 'iteratively' and can evolve as countries join and as new MDI research users and MDI launches have different data requirements
  - The MD metadata and NSI to MDI concordances allow live updates of the MDI data documentation
- Mapping units of 'firms', enterprises, legal units requires knowledge of NSI source data: registers, (weighted) sampling, sample designs
- Harmonizing variable definitions and nomenclature is done through renaming, revaluing or combining NSI variables.
  - In the `*NSI*\_MD\_conc.csv` file, information is available to show how an MD variable (from a particular MD dataset) is generated from NSI variables, through the harmonization operations remap, revalue, or redefine.
- Harmonizing values of classification variables is done by reclassifying values over time to MD standard
  - A concordance for each NSI classification version to the MDI standard is needed. Each observed value of the classification code in rawdata needs to be mapped to the MD classification, otherwise the raw data observations are lost. This is done using the concordance file `*NSI*_*classname*_MD_*classname*_classconc.csv`
- Harmonizing categorical variables is done by recoding between conforming values from codebooks
  - To harmonize data values for categorical variables, a concordance is made between `*NSI*__MD_codeconc.csv`.
- To concord other data values ((currency) units, date values), R functions are used to revalue
  - E.g. If the domains of the variable in NSI data is 1000 and in MD data 1, then the NSI value is multiplied by 1000. If the NSI value is in an R date-value, say `%d%m%Y`, an R date function is used convert to the required R date-value.

1. Concordance file – *NSI*__MD__conc.csv

This file contains the list of all variables in a particular MD panel, with information on how to map the NSI variables from one or more raw datafile (often one for each year) to the MD variable. The related table has the following column names:

```
[1] NSI_dataset,year,NSI_varname,MD_dataset,MD_varname,method,detail
```

where,

- `NSI_dataset` is the generic name of the data, that together with `year` specify the NSI datafile that hosts the variable to be use in concording. If year is empty, the concordance does not change over the years.
- `year` is the year for which the concordance holds. If empty, the same concordance rows are used for all NSI datafiles associated with the generic `NSI_dataset`.
- `NSI_varname` is the name of the variable in the NSI datafile
- `MD_dataset` is the name of the MD firm-level panel dataset, ie BR, SBS, etc.
- `MD_varname` is the name of the variable in the MD data source to be generated
- `method` is the method used to harmonize the data. The value of the categorical variable, provides the method for generating the harmonized variable `MD_varname`.
  - `revalue` The values of the variable are changed using an R function and parameters in the column `detail` and possibly from the `class` and `domains` variable from the relevant `_varnames` files.
  - `recode` The values of the variable are changed using a codebook concordance, whose name is given in de `details` column, e.g. 'NSI_filename_MD_dataset_codeconc.csv'. Only values that need to be changed require a row in the `_codeconc`.
  - `reclass` The values of the variable are changed using a classification concordance, whose name is given in the `details` column, 'NSI_classname_MD_classname_classconc.csv'. This is used to reclassify e.g. industry, region classifications.
  - `remap` The name of the variable is changed, in a one-to-one mapping from `NSI_Varname` to `MD_varname`.
  - `redefine` The MD variable is generated as a linear combination of the NSI variable. The column detail specifies the linear combination, i.e '+' or '-') in the many-to-one NSI_varname to MD_varname mapping.
- `detail` contains the function for revalue, the concordance for codebook or classification for recode and remap, and the linear operations for redefine. For revalue, any valid operation operating on the NSI_varname (referred to as `x`) is good. If the domains of the variable in NSI data is 1000 and in MD data 1, then the NSI value is multiplied by 1000, so detail = `x*1000`. If the NSI value is in an R date-value, say `%d%m%Y`, an R date function is used convert to the required R date-value, format(as.Date(x,"%d%m%Y"),"%Y")
- `NSI_datafile` is the name of the raw dataset from where the `NSI_variable` is taken from
- `year` is the reference year for that specific row, which will be used to construct the MD-cross section for that year

An example of the table for a few variables needed for Slovenian harmonized MD `BR` for year 2007 (columns `year` and `NSI_datafile` are omitted):

```
Warning in `[.data.table`(as.data.table(fread("SI_MD_conc.csv")), year == :
column(s) not removed because not found: [NSI_datafile]
```

NSI_dataset	NSI_varname	MD_dataset	MD_varname	method	detail
MIKRO_PRS_razST	MS10_razST	BR	firmid	remap	
MIKRO_PRS_razST	MS10_IZP_MS7_razST	BR	entgrp	remap	
MIKRO_PRS_razST	datum_prv_vnosa	BR	birthyr	revalue	as.Date(as.character(x), '%d%m%Y')
MIKRO_PRS_razST	datum_izbrisa	BR	exityr	revalue	as.Date(as.character(x), '%d%m%Y')
MIKRO_PRS_razST	SKD	BR	nace	remap	
MIKRO_PRS_razST	vrsta_lastnine	BR	soe	recode	SI_MD_codeconc
MIKRO_PRS_razST	datum_prv_vnosa	BR	birthyr	remap	
MIKRO_PRS_razST	datum_izbrisa	BR	exityr	remap	

NSI_dataset	NSI_varname	MD_dataset	MD_varname	method	detail
MIKRO_PRS_raw_ESTA	VAR_lastnine	BR	soe	remap	

2. *NSI*_MD_codeconc.csv

[`NSI_dataset`, `year`, `NSI_varname`, `MD_varname`, `left` , `right`]

where,

- `NSI_dataset` is the generic name of the data, that together with `year` specify the NSI datafile that hosts the variable to be use in concording. If year is empty, the concordance does not change over the years.
- `year` is the year for which the concordance holds. If empty, the same concordance rows are used for all NSI datafiles associated with the generic `NSI_dataset`.
- `NSI_varname` is the name of the variable of the specific NSI datafile associated with dataset and year.
- `MD_varname` is the name of the corresponding MD variable
- `left` gives the valid values of the categorical variable in the raw NSI dataset
- `right` gives the corresponding MDI dataset values to map

Below an example of the codebook concordance table for four variable in Portugal:

NSI_dataset	NSI_varname	year	MD_dataset	MD_varname	left	right
ifats	imputeifats	NA	IFATS	imputed	1	0
ifats	imputeifats	NA	IFATS	imputed	2	1
itgs	exim	NA	ITGS	exim	1	0
itgs	exim	NA	ITGS	exim	2	1
itgs	imputeitgs	NA	ITGS	imputed	1	0
itgs	imputeitgs	NA	ITGS	imputed	2	1
sbs	imputesbs	NA	SBS	imputed	1	0
sbs	imputesbs	NA	SBS	imputed	2	1
ictec	ICT_Type	NA	SBS	imputed	1	0
ictec	ICT_Type	NA	SBS	imputed	2	1
prodcom	unit	NA	PRODCOM	unit	kg	1 kg
prodcom	unit	NA	PRODCOM	unit	kg act sub	1 kg act. subst.
prodcom	unit	NA	PRODCOM	unit	kg N	1 kg N
prodcom	unit	NA	PRODCOM	unit	kg P2O5	1 kg P2O5
prodcom	unit	NA	PRODCOM	unit	L	1 l
prodcom	unit	NA	PRODCOM	unit	L alc 100 percent	1 l alc 100%
prodcom	unit	NA	PRODCOM	unit	m	1 m
prodcom	unit	NA	PRODCOM	unit	m2	1 m2
prodcom	unit	NA	PRODCOM	unit	m3	1 m3
prodcom	unit	NA	PRODCOM	unit	pair	1 pair
prodcom	unit	NA	PRODCOM	unit	number	number
prodcom	unit	NA	PRODCOM	unit	g	1 g
prodcom	unit	NA	PRODCOM	unit	km	1 km

prodcom	unit	NA	PRODCOM	unit	Tj	1 TJ
prodcom	unit	NA	PRODCOM	unit	kg 90 percent sdt	1 kg 90% dry

3. *NSI__classname__*MD*__classname__*classconc.csv

`[1] left,right`

where,

- `left` is the name of the current NSI classification code list for variable `NSI\_classname`
- `right` is the name of the MD classification code list the user wants to concord to

Below a sample of the Nace R1.1 to Nace R2.2 classification concordance

left	right
1422	812
1412	990
1930	3230
111	164
1310	990
1450	3821
1752	1394
150	9499
1830	1420
113	1102

Disclosure Criteria and Adherence

What are Disclosure Criteria

- The disclosure criteria in National Statistical Institutes (NSIs) are rules designed to ensure that outputs generated from sensitive firm-level data do not compromise the confidentiality of individual firms. These criteria prevent the identification of specific firms or their sensitive information from aggregated data. They are essential for maintaining compliance with privacy and data protection regulations in each country.

Components of Disclosure Criteria in the MDI:

1. Dominance Share (**domPerc**): The maximum share of the total (e.g., employment, sales) contributed by the largest 'X' firms (number 'X' defined by domNr) in a cell. Example: If domPerc is 0.75, The top 'X' firms cannot contribute more than 75% to the cell's total.
2. Minimum Number of Observations (**NumObs**): The minimum number of firms required in a cell for it to be included in the output. Example: If NumObs is 3, at least 3 firms must contribute to a cell.
3. Top Firms Count (**domNr**): Specifies how many top firms' shares are considered when applying the dominance criterion (domPerc). Example: If domNr is 1, the dominance is based on the largest firm; if 2, the top two firms are considered.
4. Dominance Variable (**domVar**): The variable on which the dominance criterion is applied, such as employment (emp) or sales (nq). Different NSIs may apply criteria to different variables, depending

on their legal requirements. **Note**: The domVar can be **'var'** in the countdown. If so, the domPerc is computed for all variables for which an aggregate is computed.

Disclosure Criteria in MDI Countries

Country	domPerc	NumObs	domNr	domVar
Finland (FI)	0.75	3	1	emp
France (FR)	0.85	3	1	-
Netherlands (NL)	0.5	10	1	-
Portugal (PT)	-	3	0	nq
Slovenia (SI)	0.5	5	1	-

- Some countries may have additional disclosure criteria, for eg. NL and SI need all exported variables to follow the dominance share criterion (not only emp (employment) and nq (sales)). In this case, the domVar is defined as 'var' in the countdown.
- The disclosure parameters are set in the Countdown when running the Infrastructure at an NSI, either by MDI or NSI staff.

How Disclosure Criteria Are Applied:

- MDI Tools like mdi_aggregate(), disclose(), DisclosCrit(), mdi_regress() and mdi_export() help automate the disclosure control for the researcher by enforcing disclosure criteria before the output is released, based on the disclosure parameters set in the Countdown.
- The primary disclosure (step 1) criteria necessitates obscuring of all cells that do not satisfy dominance criterion or cells that include less than 'minNrObs' observations.
- The secondary disclosure (step 2) criterion necessitates obscuring of one or more additional cells to protect those defined as primary confidential cells, by the minimum frequency rule. This entails suppressing the smallest cell if only one cell has been suppressed in step 1, only in case of totals/sums and if the parent node is available.
- For example, a cell that doesn't meet NumObs or exceeds domPerc is suppressed. Outputs violating criteria are flagged or excluded from export.

Disclosure Tools in MDI

Tool	Purpose	Use.by.Researchers	Use.Other.MDI.Tools
mdi_aggregate.R	Aggregate data with optional disclosure checks: dominance threshold ('domPerc') and minimum observations ('NumObs').	Yes	Yes ('disclosCrit', 'disclose')
mdi_regress.R	Performs regression analysis and automatically checks whether the number of observations meets the required minimum threshold for disclosure. Skips regressions that fail the check.	Yes	No
mdi_export.R	Exports datasets with optional disclosure compliance and metadata logging.	Yes	Yes ('disclose')
disclose.R	Performs primary and secondary disclosure checks.	No	No

disclosCrit.R	Adds disclosure metrics ('domPerc', 'NumObs') to datasets.	No	No

- The document listing all the MDI Tools and their functions is `MDI_Rpackage_0.2.1.pdf` in the 'MDI/docs/' folder.
- It is recommended that module writers use MDI Tools to comply with the disclosure criteria of all the countries that the module intends to run in.

MDI Staff + Module Writers

Setup for Researcher

- Module writers may or may not have access to actual firm-level data in a country
  - MDI documentation with available datasets and variables, plus descriptions, by country-year
  - Code can be tested using mock-data inside launcher run in a 'mock' environment that mimics an NSI
  - An R initialization program' (../launchpad/interactive_MDI.R) sets up the mock-data environment with all the metadata and Rtools.
- Module writers analyse 'consolidated' output
  - The MDI staff will stack the files from each country to create files for analysis by each module writer
  - An R 'initialization program', (../launchpad/interactive_MDI.R), will add Rtools that are useful for analysis (graphs, tables, cross-country panel regressions, etc) (*)

MDI Rtools

- There are five 'types' of R tools
  - Metadata: aid in generating, verifying, and manipulating metadata
  - Infra: to be used by MDI staff and launcher importing data, harmonization, disclosure checking
  - MDI: mostly for module writers, e.g. merge_datatables, regressions, aggregations, export
  - Analysis: help in analytical tasks and reporting
  - Programmer: aid in R coding
- Tools are documented using Roxygen2 and'exported as package'. Using standar R synatx ?function() will provide the documentation of a function, and clicing on the R package 'mdi' in the package tab of RStudio will bring up the list of functions inside the package.

Auxiliary Files

Deflators

The deflators are obtained by extracting data from Eurostat using the eurostat R library. These include deflators for value-added, capital depreciation, investments, GDP, and the harmonized consumer price index. The data is processed and cleaned to ensure consistency and relevance, with the base year set to 2010 (normalized to 1). The Eurostat datasets used include nama_10_a64, nama_10_a64_p5, nama_10_gdp, prc_hicp_aind, and nama_10_nfa, covering national accounts and price indices. Industry codes are mapped to 4-digit NACE Rev.2 classifications for compatibility with firm-level data, ensuring the deflators can be applied to various economic analyses. The resulting deflator file is a structured dataset with rows representing country codes (cc), industry codes (ind), and years (year). It includes various columns for deflators such as value-added (pnv), capital depreciation (pnc), GDP (pngdp), and consumer price index (pnhcpi). Additionally, it features asset-specific deflators (e.g., construction, machinery, intellectual

property) along with growth and depreciation rates, providing a detailed overview of deflators for further analysis.

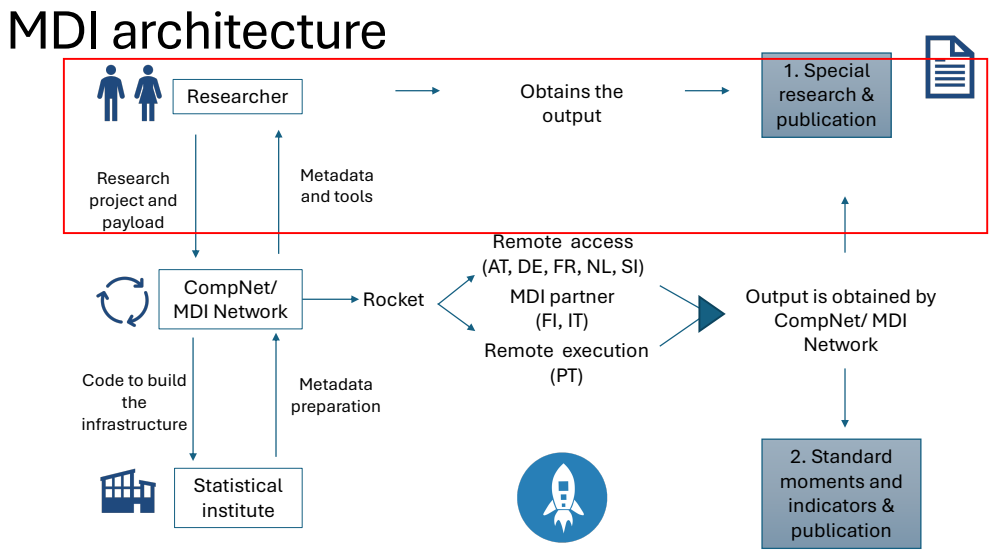
Test Server and Mockdata Environment

At present we are using the cloud server Nuvolos. This server replicates the environment of a national statistical institute and includes the MDI infrastructure. The Nuvolos server provides all the necessary tools, scripts, and libraries required to develop research modules. It also contains a mock dataset—a simulated version of real data that preserves the structure and functionality of actual data while excluding any sensitive details. More information can be found in the section "Nuvolos Server"

Module Writers (MDI Users)

This group includes productivity boards, external academic and policy researchers, and MDI 'Theme' research staff. They are responsible for creating research modules that leverage MDI's capabilities for data analysis.

Overview of MDI



Writing Modules for MDI Launcher

Writing modules for the MDI Launcher is an iterative process that evolves from conceptualization to execution. It requires a good understanding of the research objectives, data structures, and the technical capabilities of the MDI environment.

Key Steps in Module Development

- **Defining the Research Question**: Every module begins with a clear, concise research question. This question should be capable of leveraging the cross-country data available within MDI to generate insightful analyses.
- **Workflow Organization**: Divide your analysis into manageable segments. Consider general methodology; data selection; design output from NSIs; code using the MDI tools and launcher; plan cross-country analysis
  - **Data Selection**: Utilize the MDI data catalog(EB: link to rcode to MD_datafile and MD_varnames files) to select the most relevant datasets and variables for your analysis.
  - **Designing Outputs**: Plan the outputs of your module carefully, including the datasets, records, and variables to be exported from each NSI.
  - **Analysis**: Use MDI R-packages and Rtools: See the R-package libraries currently installed at NSIs and loaded at runtime of the launcher (link to file "../rocket/Rtools/Rpackages/record_package_info.csv"). If you need another package, notify MDI-staff to have the package added to NSI requirements. Use standardized functions from the MDI Rtools (link to directory "../rocket/Rtools/R"), especially when preparing output.
  - **Export data**: Using Rtool mdi_export(), generate output in each NSI, to go through disclosure checks. Collect cross-country consolidated output from TEAMS directory for module writers
- **Documentation and Summary**: Before coding, document your module's aim, the datasets and variables required, and any additional resources needed that are not part of the standard MDI environment. (EB: Link to example of module heading)

Practical module writing

- Generally two 'types' of output
  - Moments and joint moments by using mdi_aggregate()
  - Analytical output: e.g. regression results using mdi_regress()
  - Discuss: what other types are needed?
- Be aware of output size: don't get close to disclosure problems
- Output generated by mdi_export() Other output files are not exported from NSI
- Do not use code that calls 'operating system', creates (sub)directories, installs Rpackages
- Be aware of RAM limitations: be a smart coder and clean up unneeded datasets

Importing Data

- NSI metadata has all needed concordances to make the microdata panels (MD) to standards: The Launcher harmonizes and stores each MD panel. The choices are made in the file ../launchpad/countdown.R (dirTMPSAVE gives file path for saved MD panels, and flags ImportFlag (TRUE), SaveFiles(TRUE) and cleanTMP (TRUE) set logic for rocket.
- The module generated an names_select_Module file (which also was provided to the launcher to generate all the data for all the modules.
  - When reading in a file, ie BR, only read those variables that you selected
  - Double check that your selected variables are indeed available in that country. If not, provide logic in program skipping some analysis or providing and alternative.
- When merging data from two files use mdi_mergeDTs()
  - Only keep needed observations of each and needed variables (see R Joins in session 2)

Manipulating Data

- You are free to manipulate the linked panels in any way, using R

- R data.table is efficient for large datasets. dplyr and tidyverse are convenient syntaxes for many applications
- Since you may not be able to export graphs (we need to check disclosure rules in each site) export a file with enough observations of needed variables to generate display outside using consolidated data
- Hierachies of classifications are very handy to generate ad-hoc aggregations. So, instead of the standard aggregation to 2-digit or 1-digit industries, generate classifications of firms, for example by use of technology and export-status, or by magnitude of mark-ups and rent-sharing. See hier_apply() calling mdi_aggregate().
- There are tools for PIM, pim_capital(), and prod estimation, estimate_prod()

Aggregating and exporting

- Use *mdi_aggregate()* with disclosure=TRUE.
  - Stack aggregate output if they share enough columns and have rows that identify them uniquely.
- Use mdi_regress() to run models.
  - can call various regression functions (fixest package). We want to add plm.
  - generates a row of a datatable for each model that is run
  - checks for underlying number of observations for disclosure flag (especially if run 'by group')
  - Stack regression output if they share enough columns and have rows that identify then uniquely.
- Call `mdi\_export` for output from `mdi\_aggregate` or `mdi\_regress`
- Think of other types of output to generate, check disclosure, and export.

Analysis of MDI Module Output

- NSIs will upload each country's output into their Teams folder
- MDI staff will consolidate (stack) each modules outputs by country and place the module (and general launcher) output in the appropriate Teams fold for modules writers.
- The researcher can run ../launchpad/interactive_MDI.R to set up their MDI environment in a standard manner.
- The researcher analyses the output, possibly using standardized tools for statistical analysis, graphing, reporting.

Stacking metadata output files

- Among the possible files that can be exported from a country's environment there are metadata files.
- In that case, it can be that the number of files is quite large, making it much slower – and likely more expensive too – for the NSI to carry out the export process.
- For this reason, such metadata files are often stacked into broader aggregations, such as by `MD_dataset` for the `*NSI*\_varnames.csv` files.
- The following R code provides a way to *unpack* those aggregated metadata files to make sure they are stored in the proper structure.

```
'

This tool unpacks the (NSI)_(MD_dataset)_varnames.csv files into NSI_dataset-specific varname files.


'

################################################################################
```

```
MD_datafiles <- fread(paste0(dirPAYLOAD,'MDmetadata/MD_datafiles.csv'))
MD_varnames <- fread(paste0(dirPAYLOAD,'MDmetadata/MD_varnames.csv'))

files <- list.files(paste0(dirROCKET,'NSImetadata/',CountryCode,'/'), recursive = T)

files <- files[grepl('_varnames',files)]

for (file in files) {
  data <- fread(paste0(dirROCKET,'NSImetadata/',CountryCode,'/',file))

  NSI_files <- unique(data$filename)

  for (NSI_file in NSI_files) {
    data_filt <- data[filename == NSI_file,]
    data_filt$filename <- NULL
    names(data_filt) <- c('NSI_varname','is_key','description','class','domain')
    fwrite(data_filt, paste0(dirROCKET,'NSImetadata/',CountryCode,'/',CountryCode,'_',NSI_file,'_varnames.
  }
}
```

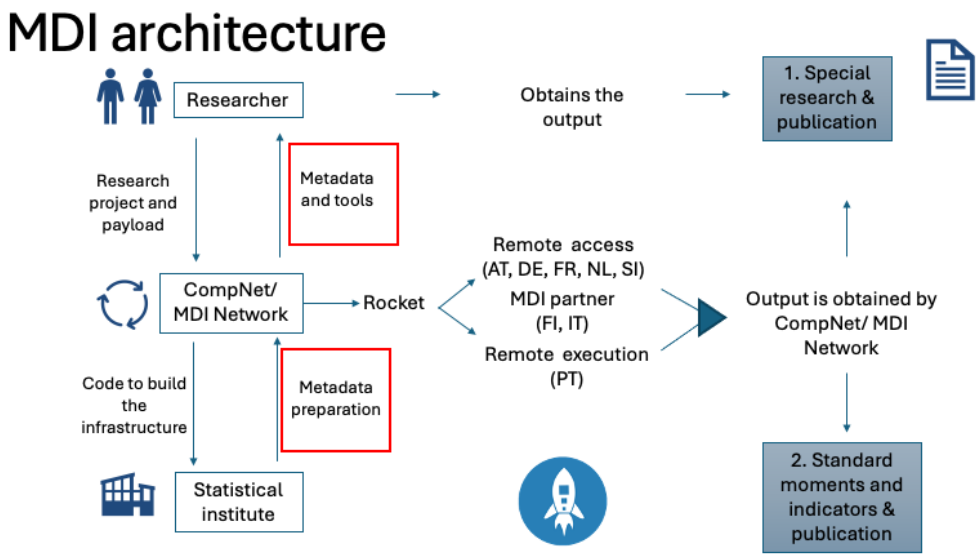
Work at NSI

Some more details of steps at NSI

- Bring MDI directory structure into NSI environment, including Rtools, metadata etc
- Install the needed R packages
- Define root of path to data files
- Place list of all files, with info, into Metadata directory
- Place lists for each 'classification' variables into Metadata directory
- Place list of all variables, by filename, type, and codebook names in Metadata directory
- Provide 'concordances' for classifications and for variable to map to 'MDI' version

Preparation for each Launch

- Fill in country specific parameters in `launchpad\countdown.R`
- Once for each launch, run `prepare_NSI.r` for generating new metadata
- Once for each launch run `liftoff.R`, with countdown.R parameter SaveFiles<-TRUE and cleanTMP<-FALSE. This makes and stores harmonized MD files for use with payload. Keep the temporary files on disk during full launch window, unless re-run is needed
- Run `liftoff.R` , with countdown.R parameter ImportFlag<-FALSE, so that payload modules will use the above temporary MD files. (if there are fixes to be made to the data, run again with ImportFlag<-TRUE).
- Conduct disclosure checking of files - Upload output to Teams (LINK HERE to directory for a particular CC)



The Harmonzied MD Longitudinal Firm Panels

- see **?@sec-nsi**
- In NL and France, NSI harmonized the raw datafiles to look like MD datasets. Not much concording done by launcher.
- We want to shift the boundary between work of NSI and MDI
  - NSIs document their files: Launcher reads files and uses metadata to generate MD panels.
  - NSI does all harmonization to MD standards: Launcher reads the files into R
- Advantage to having Launcher do the work:
  - Reduce maintenance costs for NSI (annual updates)
  - Improve codification of conceptual work done by NSI staff
  - …but higher cost to learn rigid method of documentation/codification of metadata and to coordinate with MDI

MDI Infra Tasks

Preparation for Launch and Mission Control

- Pre-launch testing
- Lockdown of version, transfer to TEAMS/NSIs
- Interacting NSI and MDI staff during launch period
- Collecting output

Pre-launch Testing

- Use `mdi.labs.vu.nl` site, with rs-user login
- Start with fresh R each time (under options, choose to never save and datasets, open files, loaded packages; Quit R before quitting the remote server).

- Always test 'from the beginning'.
- Test using different countdown parameters, on dataset saving, tempdir, disclosure. (discuss with mockdata folks to make data to allow for different types of tests).
- Have a formal testing 'checklist' and have a sub-team approve the release

Lockdown of version

- Lock the R package list
- Make Github branch `Launch v2.1`
- Run file/program to get github commit of version (Marco?), re-commit `Launch_v2.1`
- Run scripts to transfer from github to NSI specific Teams site
- Ask NSIs to download. Make a new MDI root directory voor v2.1 (back-up the old one). ave NSI sysadmin update required R packages, and roxygenize and install the MDI package.
-

Interaction NSI - MDI during launch

- download zip of MDI to `dirMDI`, keep zip file.
- Edit MDI metadata and progs as needed.
  - Flag if change pertains to other countries. Move up chain, and back to NSIs with fixes
  - In extreme case, decide to relaunch…..
- We write script using R file-info to compare MDI and MDIzip after launch is finished. NSIs send metadata fixes to us, tools fixes are reported…

Work on MDI tools

- Metadata tools: list2hier, hier2cp
- MDI tools
  - improved harmonizing and concording
  - traversing hierarchy
  - index number aggregation
- Infra tools: mock data generation and environment
- Analysis tools: What do we want?

Mockdata

- Mock data is key
  - Code development for Rocket needs to be tested
  - New MDI staff need to be trained
  - Country metadata and concordances need testing, requiring country specific mock data
- Specifications for mock data
  - All files from NSI_datafiles.csv are covered, with all years as given by NSI_filename_varnames
  - All varnames from all NSI files, with proper format and domain (including classification, codebook etc)
- Underlying 'firm' datasets
  - SBS/BS cobb douglas, deterministic with stochastic draws. Start with firm size and back out k, m, y from stochastic productivity and k/l moments
  - SBS/BS forward looking Hopenhayn with stochastic prod draw and shocks to prod and demand: endogenous exit.
  - SBS/BS Firms a la Aglio, Bartelsmanj with draws of A/g and η, ρ.
  - Firm dynamics with innovation, extensive choice of innovative activity (ICTEC, R&D)

  - Firms with extensive and intensive trade choices (ITGS, ITS, OFATS, IFATS)

break tasks into small progs (scripts) and tools (functions)

- Tools to read country 'metadata'. Not only datafiles, varnames, but also CompNet stuff on number of firms, prod distribution:
  - For a compnet country/industry pair: what is a good parametric prod distribution? (gamma 2,3). Target proper mean/median salers per worker.
  - distribution of number of firms per industry in a year
  - average entry/exit rates of firms
  - average entrant size
  - exit hazard by age and size
  - trade hazard by firm productivity
  - intangible hazard by firm productivity
  - credit constraint hazard by firm productivity
- For items missig for now, build this table with made up numbers, as default
- draw 'universe' of firmd identifiers and enterprise identifiers with concordance
  - use metadata from country to understand share of firms that are not 1-to-1 with group, and median number of firms in group with more than 1 firm.

Workflow for mock data simulation

- BLOCK0: Prepare Auxiliary Files
  - Define the country, the sample periods, the datasets and read country-specific NSI metadata (datafile, varname, codebook).
  - Create a table specifying the hierarchical structure among variables.
  - Develop a table defining the concordance between fundamental model variables and NSI variables.
  - Compile a file detailing auxiliary regressions for predicting numerical, logical, and categorical variables.
- BLOCK0: Obtain Data Moments from the Data or by Simulation
  - Calculate the sample mean and variance of employment for each NACE 2-digit sector.
  - Determine the average exit rate for each NACE 2-digit sector.
  - Extract regression coefficients for auxiliary regressions.
  - Gather information on sample sizes of surveys.
  - Compute key economic ratios and rates: capital-labor ratio, capital rental rate (interest rate), wage rate, and capital depreciation rate.
- BLOCK1: Simulate an Unbalanced Panel Dataset
  - Generate an unbalanced panel dataset for firms over time, incorporating firm entry and exit dynamics based on Hopenhayn (1992).
  - Estimate model parameters: α (output elasticity of labor), σ (standard deviation of TFP process), and `z_exit` (exit threshold for firms' productivity) by targeting the sample mean and variance of the firm size (employment) distribution and the exit probability.
  - The simulated panel data includes firm ID, year, productivity, labor, capital, depreciation, and EBITDA.
- BLOCK2: Predict BR and BS Variables
  - Use concordance tables between model variables and NSI variables, as well as auxiliary regressions and regression coefficients to predict BR and BS variables.
- BLOCK2: Sample from the 'Universe' of Firms

  - For each survey table, sample from the firm universe and predict NSI variables using the auxiliary regressions and regression coefficients.

Nuvolos Server

Nuvolos is the server environment that we use for trainings, testing, developing and debugging. There are two seperate environments:

1. MDI Training Environment
2. MDI Internal testing Environment

If you want to get access to either of the environments, reach out to Johanna via email or Teams.

MDI Training Environment

This space is designed for training, practicing, and familiarizing with the MDI infrastructure. It's intended mostly for external people, as it already contains all relevant files, ie. the whole MDI infrastucture, mockdata. It's an environement in which externals can test their codes using the MDI infrastructure and mockdata. It is updated periodically, so doesn't always reflect the latest version of the MDI. It is not intended for bug-fixing and working on the infrastrucutre, as it doesn't link to Github.

"Files" section

You'll find the following folders:

- MDI: containing the MDI infrastructure
- TMP: containing temporary files
- NSIdatafiles: containing mock data files imitating real data
- output: if a script generates any output it can saved here

Writing and executing code

=> go to the section "Applications" on the left navigation bar and open the RStudio application

- Run the countdown:
  - In RStudio, on the right in the files section, look for the MDI > launchpad and open countdown.
  - R Click on "source" to execute it.
  - You will be asked choose a program to run, choose interactive MDI by entering 4
  - Wait until the script is done. The metadata, Rtools and libraries are imported - you can now create/ run your script.
- Create/ Execute a Module
  - Add a new file
  - Add your module code or any code that you want to run
  - Execute your code either line-by-line by clicking "run" or all at once by clicking "source"
- When you run your script, any errors will appear in the Console section. The executed lines are highlighted in blue, while errors are displayed in red. To resolve an issue, identify the problematic line in your script, make the necessary corrections, and run it again.

Internal Testing Environment

This space is designed for writing code, creating modules and testing scripts on mock data and debugging. It is intended for the MDI team and module writers who are familar with the MDI and have access to the

MDI Github repository. Each user needs to connect their Github account in order to pull/ push changes. - Each user has their own space that means - until you push your code to Github - your changes are only visible to yourself. - You have to make sure yourself that you're working on the latest version of the MDI, ie. pull changes from Github when you open RStudio, find the most relevant commands below. - You'll find NSI mock data as well as harmonized MD mock data that you can use for testing and developing your scripts.

First time users: How to connect Github

- Navigate to applications (left menu) and open RStudio
- Go to the terminal
  - Generate a public/private key pair by executing this command: `ssh-keygen -t ed25519` (no need to change suggested location or create a password > press enter 3 times)
  - Navigate to the folder where both are saved. You can do that on the file section on the right side, you might have to click on the "/" to see all directories. The folder .ssh is a hidden folder, so click on the gear symbol and select the option "Show Hidden Files"
- Open the id_ed25519.pub and copy it's content
- Open Github in the browser
  - In Github: go to settings>SSH and GPG keys
  - Click "add new SSH key"
  - Add the copied pub key into the key section and add a title eg. "Nuvolos MDI test environment", then save
- Back to Nuvolos, in RStudio, Terminal, clone the branch using this command (from within /files folder, which is default):
  - `git clone --branch pre_Launch_v2.2 --single-branch git@github.com:Secretariat-CompNet/MDI.git`
  - The MDI with all files will show up in the files section on the right
  - Then go to home/datahub/ in the file section and open the .gitconfig file. The file will look like this:

```
[user]  email = 12345678+Name@users.noreply.github.com
      name = YourName
[credential]
      helper = cache --timeout 64800
```

- Make sure that the email address is the one from your Github, not eg. the iwh email address. To check which is the right one go to your Github account > Settings > Emails. Copy the email address ending on @users.noreply.github.com as shown below into the .gitconfig file. Save file.
- The MDI is correctly set up now. You can run your code with mockdata, edit and pull/push changes to Github
- To test codes: execute countdown and select interactive_MDI (option 4) before running your own script.

The most common git commands

- To push to/pull from Github, go to terminal
  - Make sure your in the MDI folder (change dir with this command: cd /files/MDI)

- To check changed files use: `git status`
- Add file to commit eg: `git add launchpad/countdown.R` (path to file or . if you want to add all changed files)
- Commit changes `git commit -m "Your commit message"`
- To pull changes from Github: `git pull`

names_select_Module

Back to main

mdi_mergeDTs()

```
# Merge BR and SBS by firmid and year, only selecting firms that are ever in the SBS
# DO NOT ALLOW Cartesian: firmid and year need to be the unique record descriptors; set to FALSE
#but do a left join of the selected br dataset (only firms that are ever in SBS), and the SBS
sbs_firms <- unique(sbs[,.(firmid)])
setkey(sbs_firms,firmid)
br_sbs <- br[sbs_firms, nomatch=0]
br_sbs <- sbs[br_sbs, on=.(firmid, year),nomatch=NA]
```

Back to main

NSI_initialise()

Back to main

mdi_export()

```
#| eval: true
#| echo: true
#Version : R 1.1
#
#Function to export a given data file in a specific format and add an entry to the output description file
#
#INPUTS
#output      : a data.table object
#format      : (character) csv, RDS, txt, dta, xlsx, sas
#output_name : (character) name of the file
#output_path : (character) specified directories
#desc_file   : name of the file that describes the output
#output_type : (character) either 'sum_stat' for summary statistics, 'reg_tab' for regression table or 'otl
#description : (string) allows further explanation of the output file
#hier        : (booelan) set to TRUE if there are multiple levels of aggregation (child and parent nodes)
#hhfile      : if used in mdi_aggregate_h: name of hierarchy file used for aggregation
#
#
#OUTPUT
#Objects saved in specified directories.
# Usage
```

```
# export_db(
#     output = datalist_num_firms[[nm]],
#     format = "csv",
#     output_name = paste0(CountryCode, "Coverage_Number_Firms", nm, MDIversion, sep="_"),
#     output_path = dirOUTPUT,
#     desc_file = Output_description.txt,
#     output_type = 'sum_stat',
#     description = 'Number of firms by datasource, industry (2 digit, 1 digit) and year',
#     hier = TRUE,
#     hhfile = indHier)




export_db <-
  function(output,format,output_name,output_path,desc_file,output_type="sum_stat",description=NULL,hier=FA

    # create entry in output description file
    if (output_type == "sum_stat") {

      # use attributes set in 'mdi_aggregate'
      #vars <- attr(output, "sumvars")
      #bygroups <- attr(output, "sorted")
      #newentry <-
      #  paste(output_name,": Summary Statistics of",paste(vars, collapse = ''),"by",paste(bygroups, colla
      #        description,"\n",sep = " "
      #  )
      newentry <- paste0(output_name, ": Summary Statistics: ", description, "\n")
    } else if (output_type == "reg_tab") {
      newentry <- paste0(output_name, ": Regression table: ", description, "\n")
    } else if (output_type == "other") {
      newentry <- paste0(output_name, ": ", description, "\n")
    }
    write(newentry,append = TRUE,file = paste0(output_path, desc_file, ".txt"))



    # always check disclosure if output is type summary statistic
    if (output_type == "sum_stat") {
      output <- disclose(output,hier=hier,hhfile=hhfile)
    }


    # export file
    file <- paste0(output_path, output_name, ".", format)
    if (format == "csv") {
      fwrite(output, file)
    } else if (format == "RDS") {
      saveRDS(output, file)
    } else if (format == "txt") {
```

```
    #To A Tab Delimited Text File
    write.table(output, file, sep = "\t")
  } else if (format == "dta") {
    write_dta(output, file)
  } else if (format == "xlsx") {
    write.xlsx(output, file)
  } else if (format == "sas") {
    write_sas(output, file)
  }
}
```

Back to main